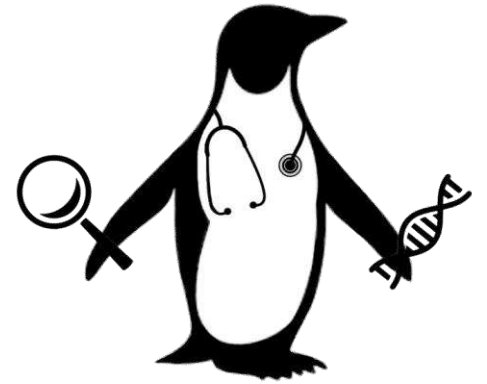




The PENGQUIN Odyssey:

A Quest for Query-able Semantic Genomic Data for Clinical Practice

Elias Crum

SIB Vital-IT Group Meeting
April 4th, 2025

SIB Research Stay: 1 March – 30 April

A Quick Thank You!



Tarcisio Mendes

Ana-Claudia Sima

Jerven Bolleman

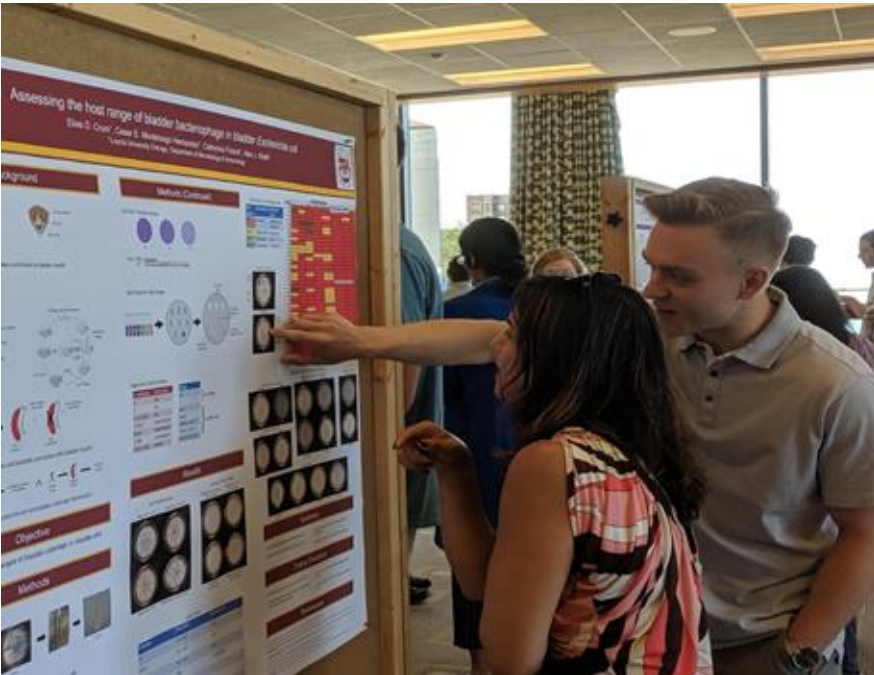
Marco Pagni

An Intro to Elias

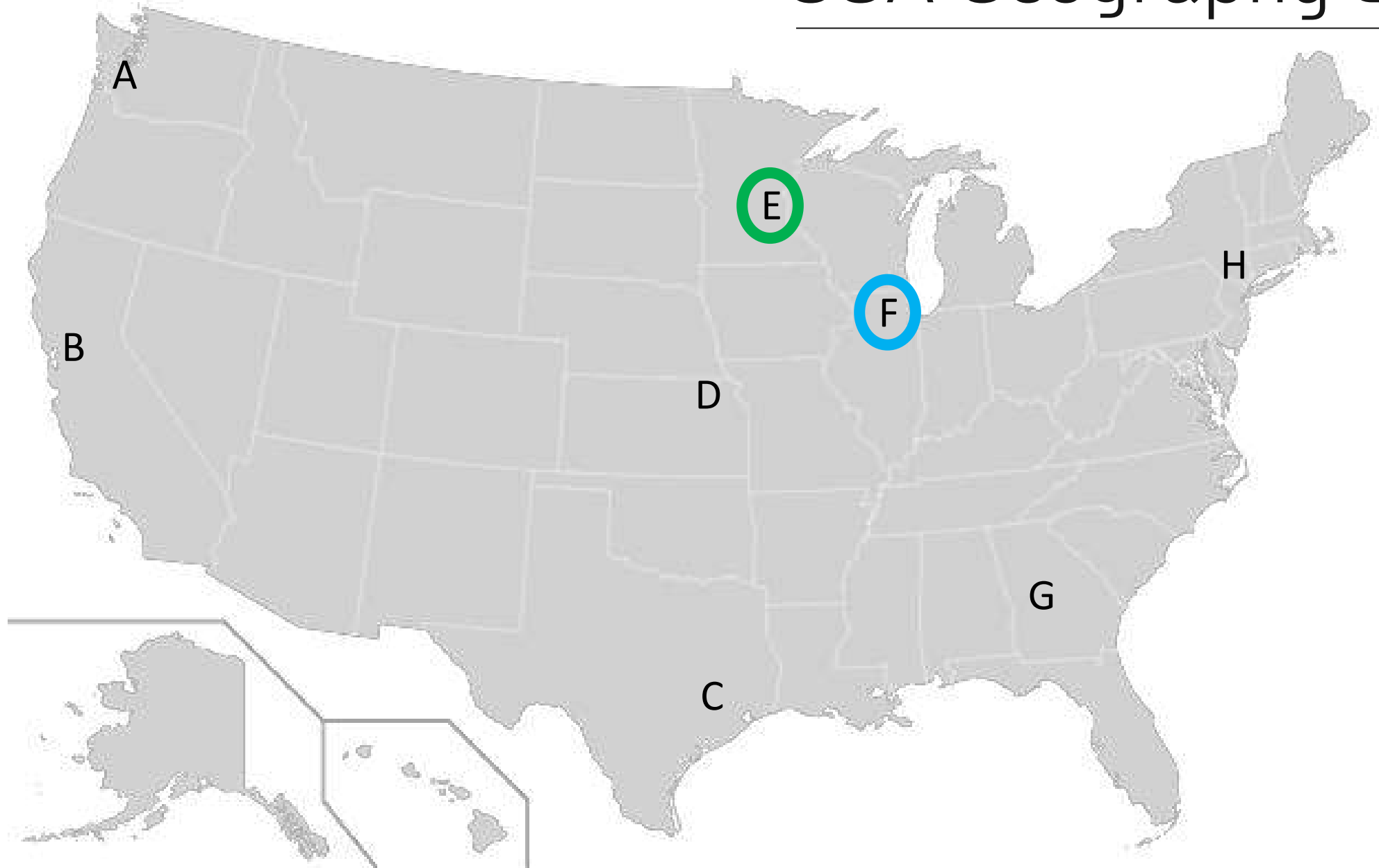
My Journey:



BSc + MSc - Bioinformatics



USA Geography Lesson

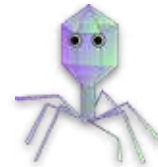


An Intro to Elias

My Journey:



BSc + MSc - Bioinformatics

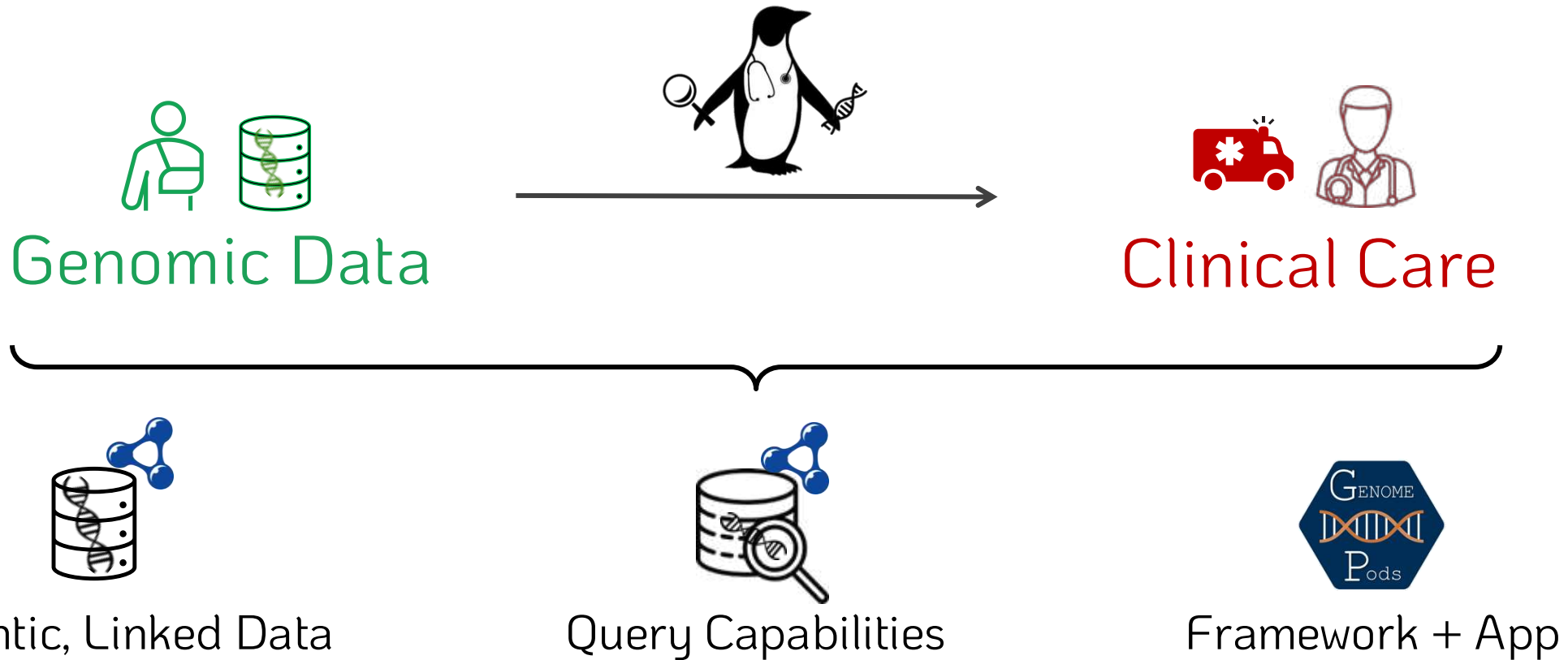


Prokaryotic Genomics

How can we increase **genomics** use in **medicine**?

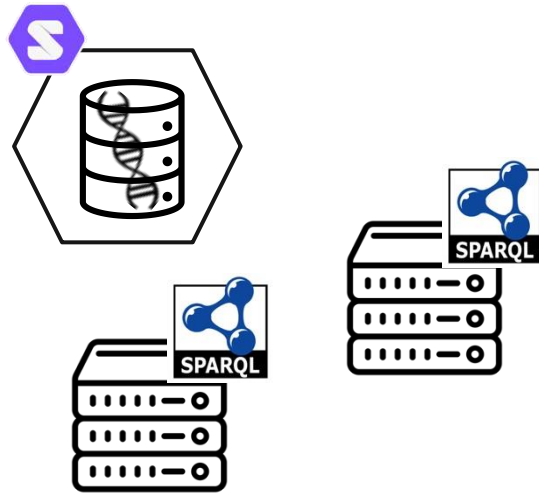
PENGQUIN:

PErsoNal Genome QUery IN
healthcare and clinical practice

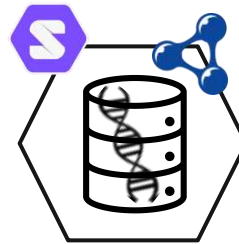


PENGQUIN Goals

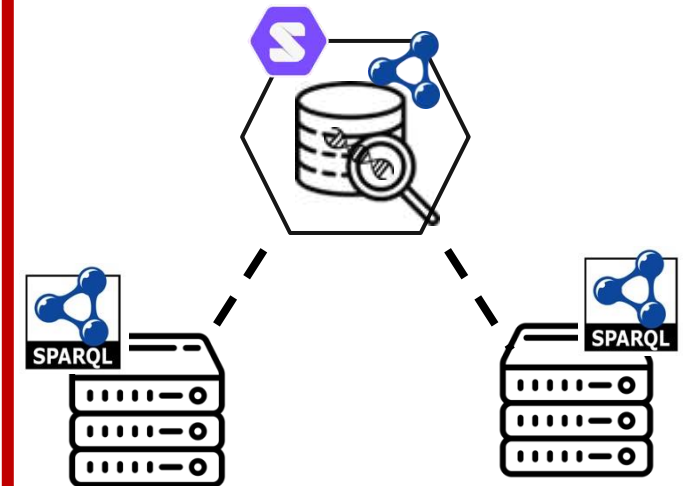
Data Storage



Genomic Data Representation



Question Answering



**Framework
& web application**



PENGQUIN Goals

Improve genomic data **USAGE** and **SCALABILITY**



Improve genomic data store accessibility (while maintaining privacy)



Represent genomic data semantically as Linked Data



Query genomic data and linkages



Connect to existing initiatives / applications

Proof-of-concept



Develop a framework and web application

Talk Outline



Decentralized Data Storage (*CHIST-ERA TRIPLE Project*)



Introduction to Solid Pods



Introduction to TRIPLE Project



The Solid Cockpit web application (and maybe a use-case example if we have time)



Genomic Variant Data Semantification (*Semantic VCF*)



Why Semantic VCF Data?



Existing vocabularies/tools



Federated Querying (*Real World Federated Query Survey*)



Motivation + Challenges

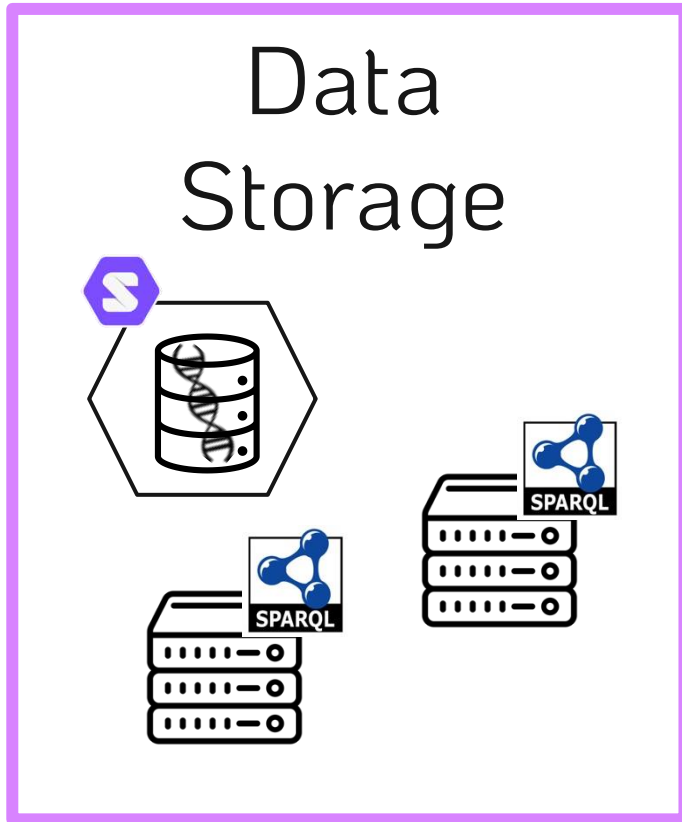


Progress and preliminary findings



Future aims

The First Act – Private Data Storage



CHIST-ERA TRIPLE Project



Swiss Institute of
Bioinformatics



GHENT
UNIVERSITY



ÚOCHB
IOCB PRAGUE



Address current **challenges** of using **RDF resources**
to bring the **utility** of these resources to a **broader group of researchers**

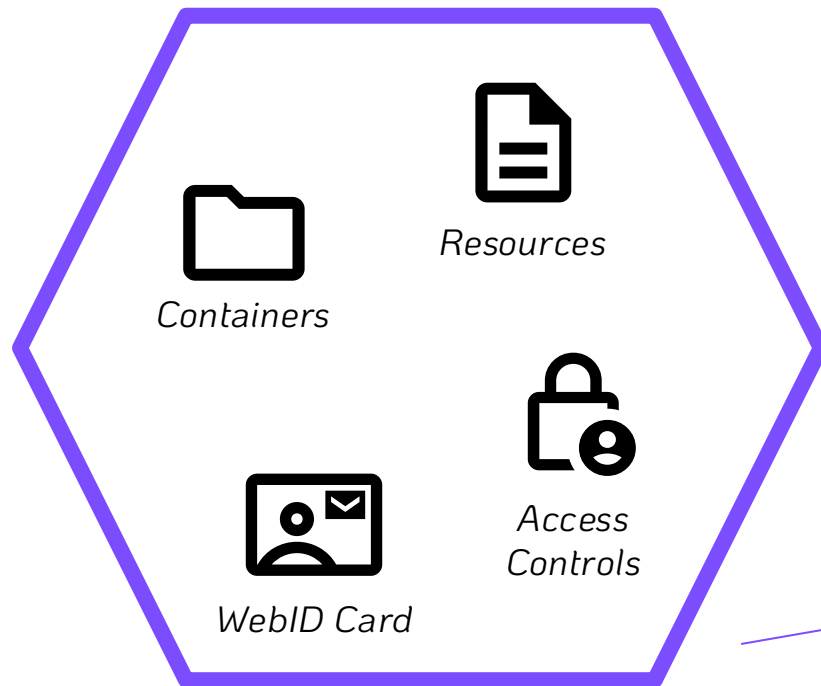
What is a Solid Pod?



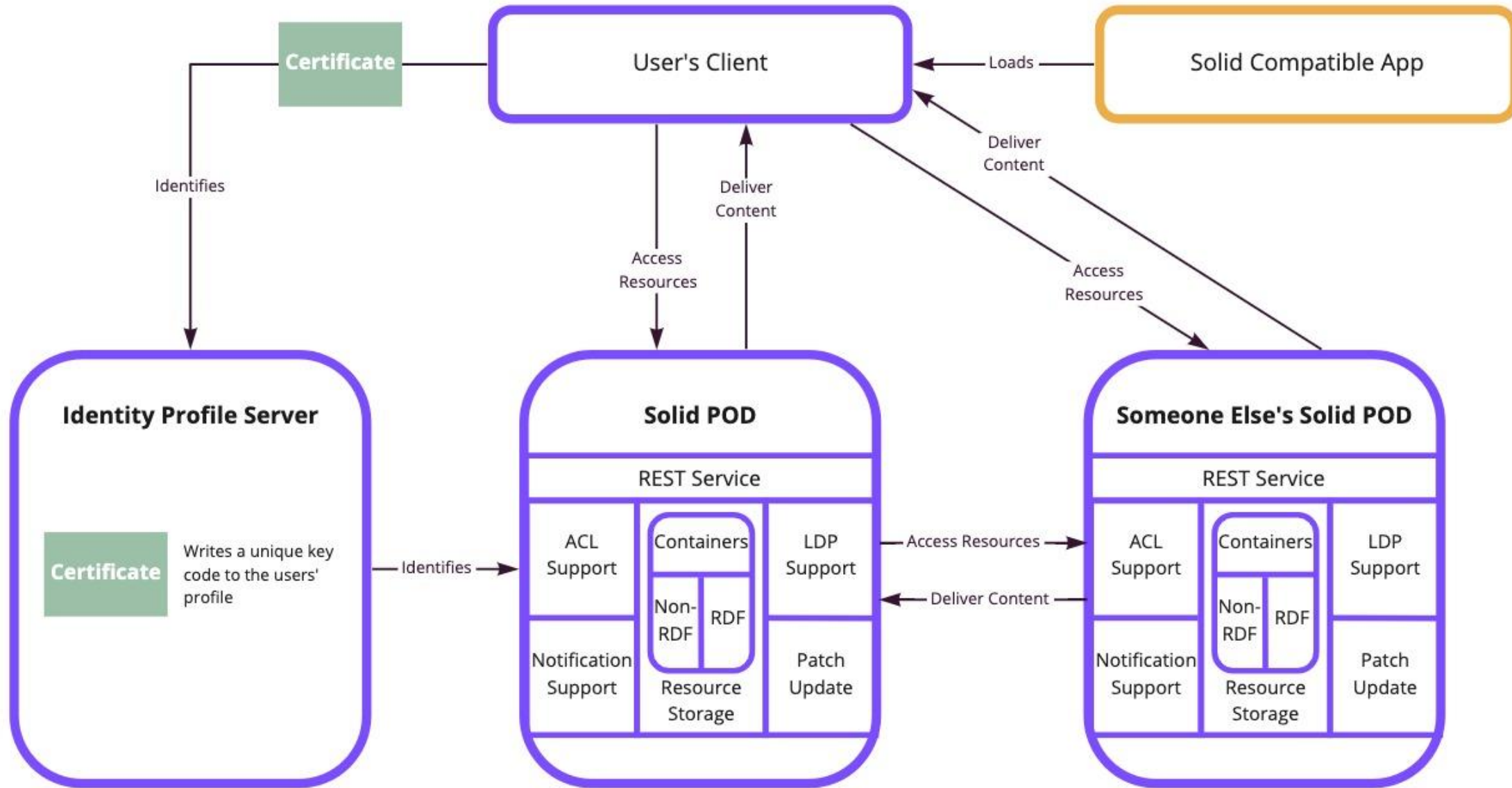
Secure, Decentralized Data Storage Specification

- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*
- ↳ *Pod exists on a Server (accessible to the Web)*

Elias' Pod



Solid's Architecture



miro

Solid Pod vs *SPARQL Endpoint*?



Solid Pod

- + decentralized storage strategy
- + can store both file data and RDF data
- + granular privacy policy capabilities
- native querying capabilities limited
- not designed for large-scale data
- no computation attached to data store



SPARQL Endpoint

- centralized storage strategy
- triple store (only RDF)
- limited privacy capabilities
- + designed for high-performance querying
- + excels with large datasets
- + computation directly attached to data store

How to query a Solid Pod?

A Query Engine



- Query computation client-side
- Allows for SPARQL query execution “over” Solid Pod API
- Linked Data traversal + heterogeneous result joining capabilities

SPARQL Query

```
1 SELECT DISTINCT ?p {  
2   ?s ?p ?o  
3 } LIMIT 10
```

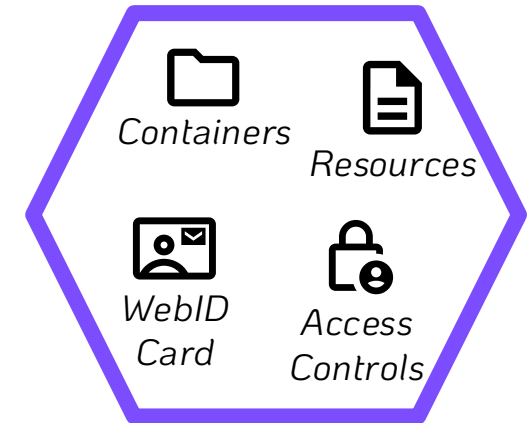
+ Source:
Elias' Pod

Results Binding Stream
SPARQL+JSON



Execute
SPARQL

Elias' Pod



Request

Download

Solid Pod Implementations?



BBC MyPDS App



Flemish Data Utility Company

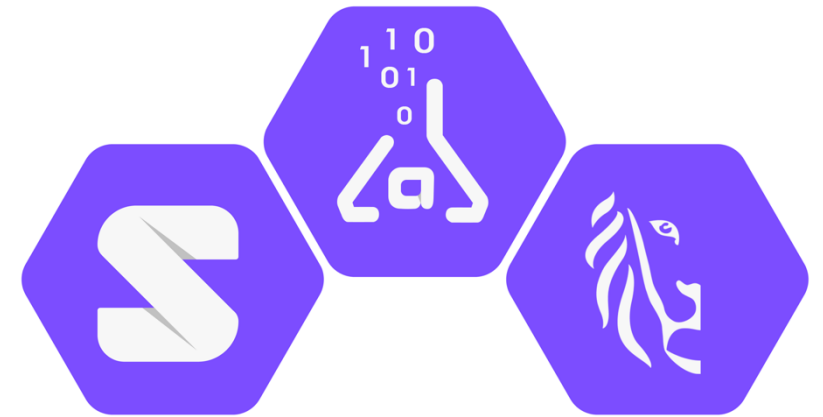


health data vaults



Flanders
State of the Art

Flemish Smart Dataspace



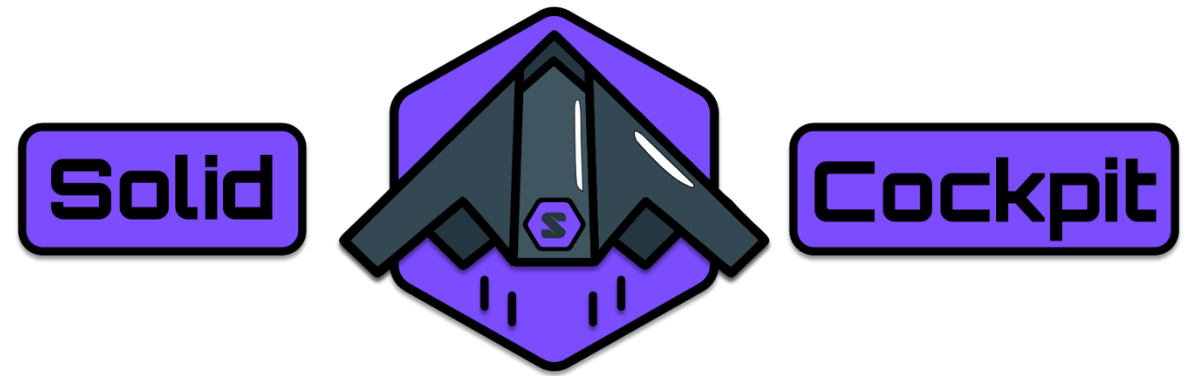
Solid Lab Use-cases

CHIST-ERA TRIIPLE Project



**Standardized documentation
and resource statistics**

My Contributions



A Web Application to:

knowledgeonwebscale.github.io/solid-cockpit/

☁️ Upload Data to a Solid Pod

🗂️ Browse contents of a Solid Pod

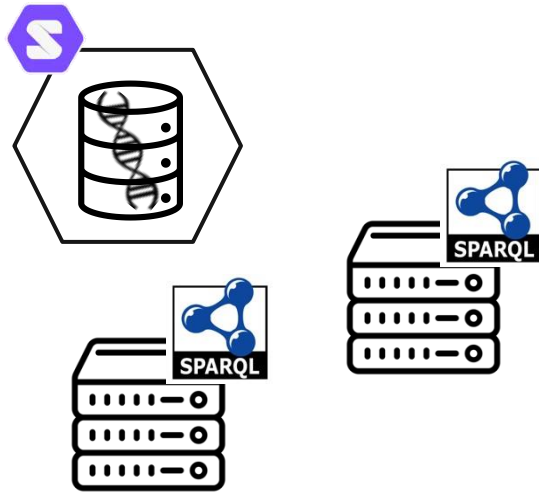
🔒 Edit Privacy of (and share) data in a Solid Pod

🔍 Query Data in Solid Pods + SPARQL Endpoints

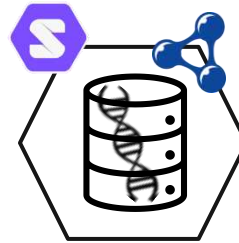
↳ Implement a query cache for storing queries and results

The Second Act – Semantifying Genomic Variant Data

Data Storage



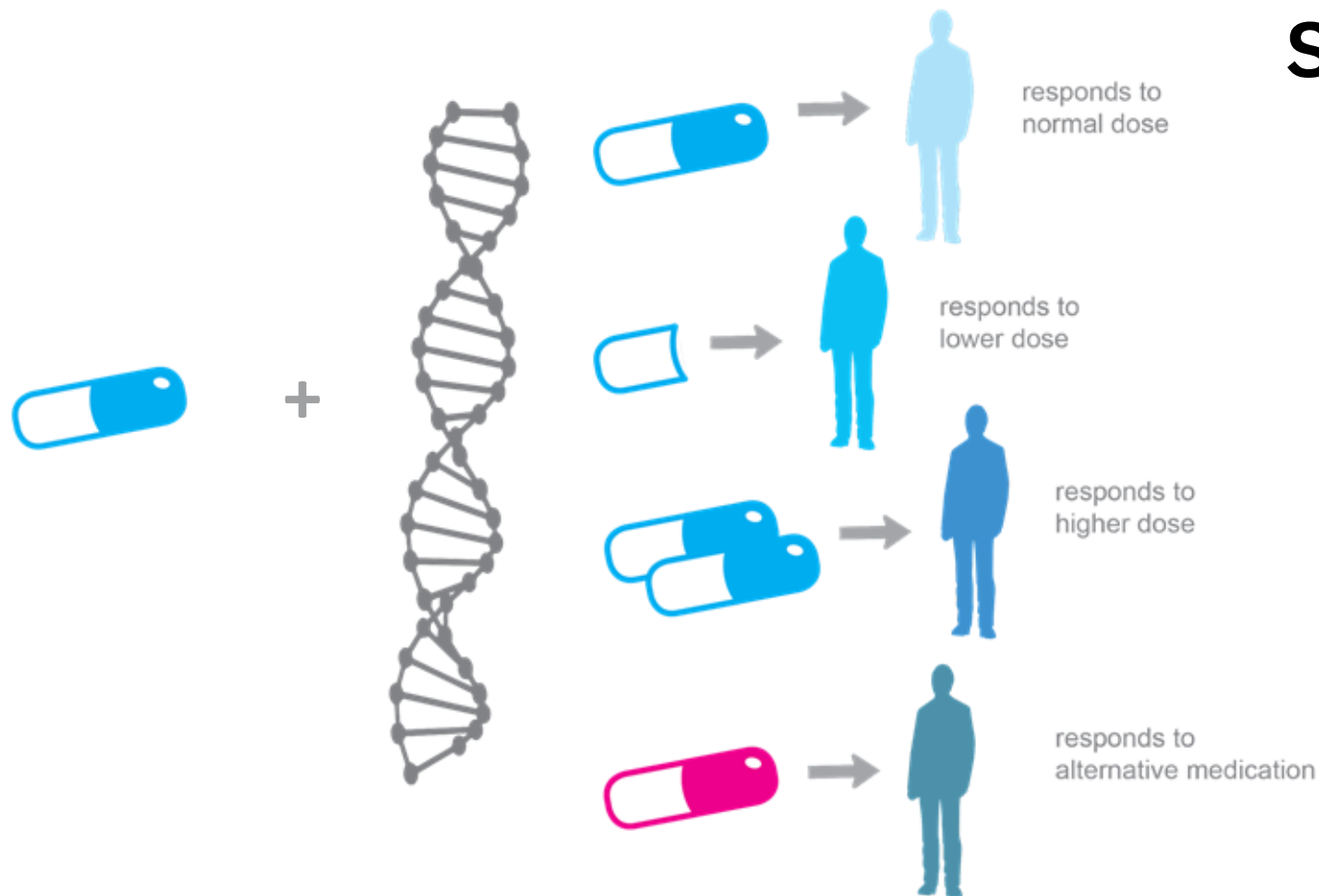
Genomic Data Representation



Representing Genomic Variant Data Semantically

Genomic Data, Health Care, and Semantics?

Pharmacogenomics Use-case (E.g. Warfarin):



Semantic Genomic Variant Data?

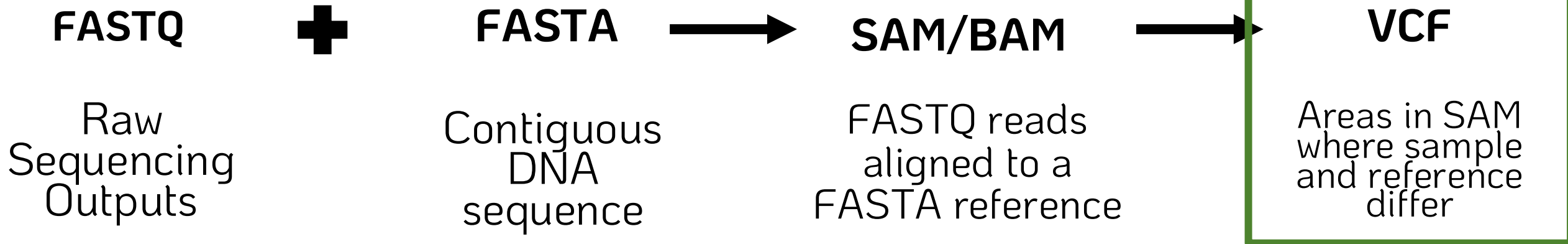
Research:

↳ **F.A.I.R.** –
Findable. Accessible.
Interoperable. Reproducible.

Clinic:

↳ Data Linking
Data Provenance
Data Interoperability

Semantic VCF – Sequencing Data Formats Intro



1	>gi 568336023 gb CM000663.2 Homo sapiens chromosome 1, GRCh	
2	CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	@HD VN:1.0 SO
3	ACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAA	@SQ SN:chr20
4	TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCT	@PG ID:TopHat
5	CTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	lign-edit-dist 2 -
6	CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCCTAACCC	20 /data/user446/m
7	CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	HWI-ST1145:74:C101
8	TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTCGCG	CCGTGTTTAAA
9	TCTGACCTGAGGAGAAGCTGTGCTCCGCCTTCAGAGTACCAC	C BBDCCDDCCDDDD
10	CGCTCTCGCGGTGCTCTCGGGTCTGTGCTGAGGAGAACGCA	AS:i:-15
11	CGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCA	HWI-ST1145:74:C101
12	AGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCC	CGTGGATCAT
13	CGCGCCGGCGCAGGCGCAGACACATGCTAGCGCGTCTGGGGTG	G DCDDDEDDDDDD
14	GCGCGCCGGCGCAGGCGCAGAGACACATGCTACCGCTCCAC	AS:i:-16
15	AGGCGCACCGCGCGCGCAGGCGCAGAGACACATGCTAGCG	
16	GCAGAGACGCAAGCCTACGGGCGGGGTTGGGGGGCGTGTG	
17	GCTGGGCGGGGGGAGGGTGGCGCCGTGCACGCGCAGAACT	
18	GGTAGAACCTCAGTAATCGAAAGACGGGATCGACCGCCCC	
19	GCTTGCTACGGTGCTGTGTCAGGGCGCCCTGTGCGGAC	
20	GAGTGGTGGCCAGCGCCCCCTGCTGGCGCCGGGCACTGCAG	

@FORJUSP02
CCGTCAATTC
+
AAAAAAAAAA

Q scores (as ASCII ch

Base=T, Q=':'=25

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
```

```
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
```

FORMAT	NA00001	NA00002	NA00003
GT:GQ:DP:HQ	0 0:48:1:51,51	1 0:48:8:51,51	1/1:43:5:.,.
GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3	0/0:41:3
GT:GQ:DP:HQ	1 2:21:6:23,27	2 1:2:0:18,2	2/2:35:4
GT:GQ:DP:HQ	0 0:54:7:56,60	0 0:48:4:51,51	0/0:61:2
GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3

```
@HD      VN:1.0  SO
@SQ      SN:chr20
@PG      ID:TopHat
lign-edit-dist 2 -
20 /data/user446/m
HWI-ST1145:74:C101
      CCGTGTTTAAA
C      BBDDCCDDCCDDDD
      AS:i:-15
HWI-ST1145:74:C101
      TGTGGATCAT
G      DCDDDEDDDDDD
      AS:i:-16
```

```

HWI-ST1145:74:C101DACXX:7:1204:14760:4030      16      chr20      270877      50      100M      *      0      0
      GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C      DDDDDDDDDCCDDDDDDDDDEEEEEEEFFFEFFEGHHHFGDJJIHJJIJJJJIIIGGFJJJIHIIJJJJJJIGHHFAHGFHJHFGGHHFFDD@BB
      AS:i:-11      XM:i:2      XO:i:0      XG:i:0      MD:Z:0A85G13      NM:i:2      NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699      0      chr20      271218      50      50M4700N50M      *      0
      0      GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCCTTGGGTCTCCGAGCAGAACATCCTCAAATATGACCTCTCG
accepted hits.sam

```

What does a VCF File Look like?

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0 1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1 0:77	1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

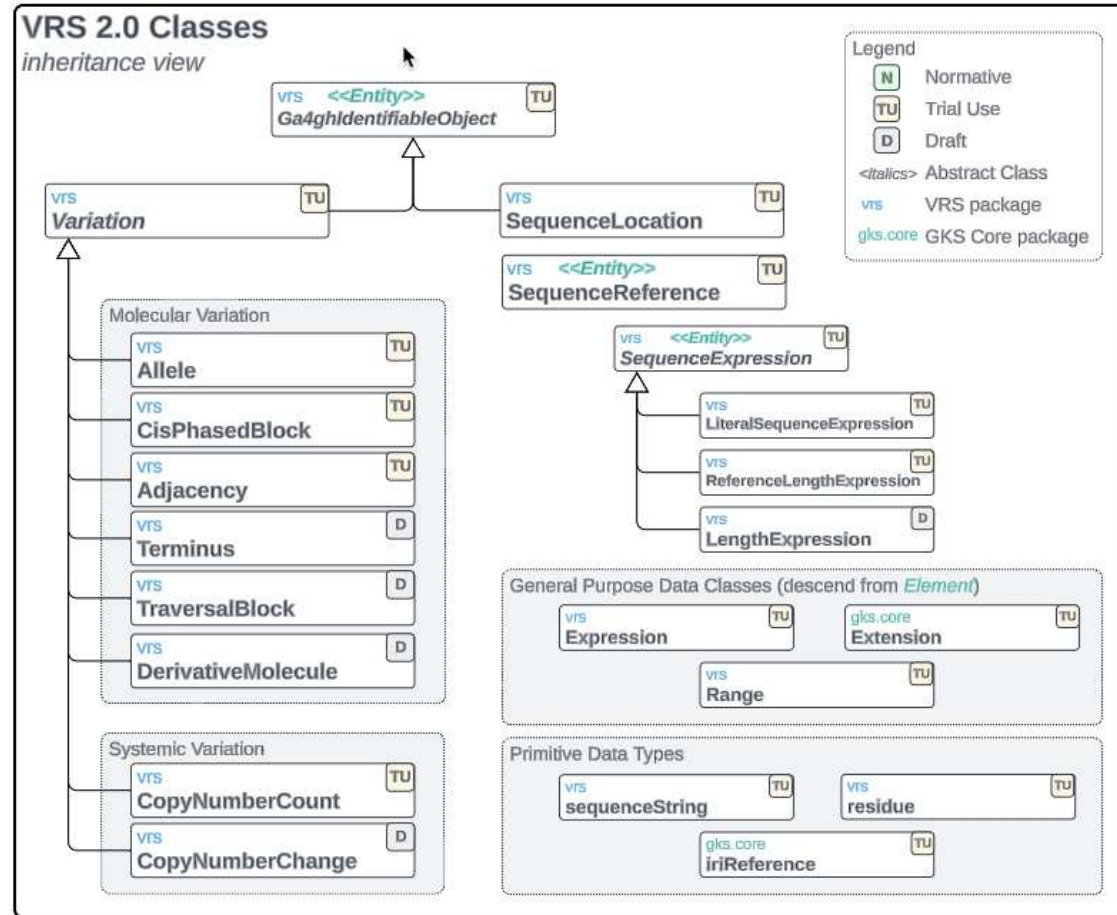
Insertion

Other event

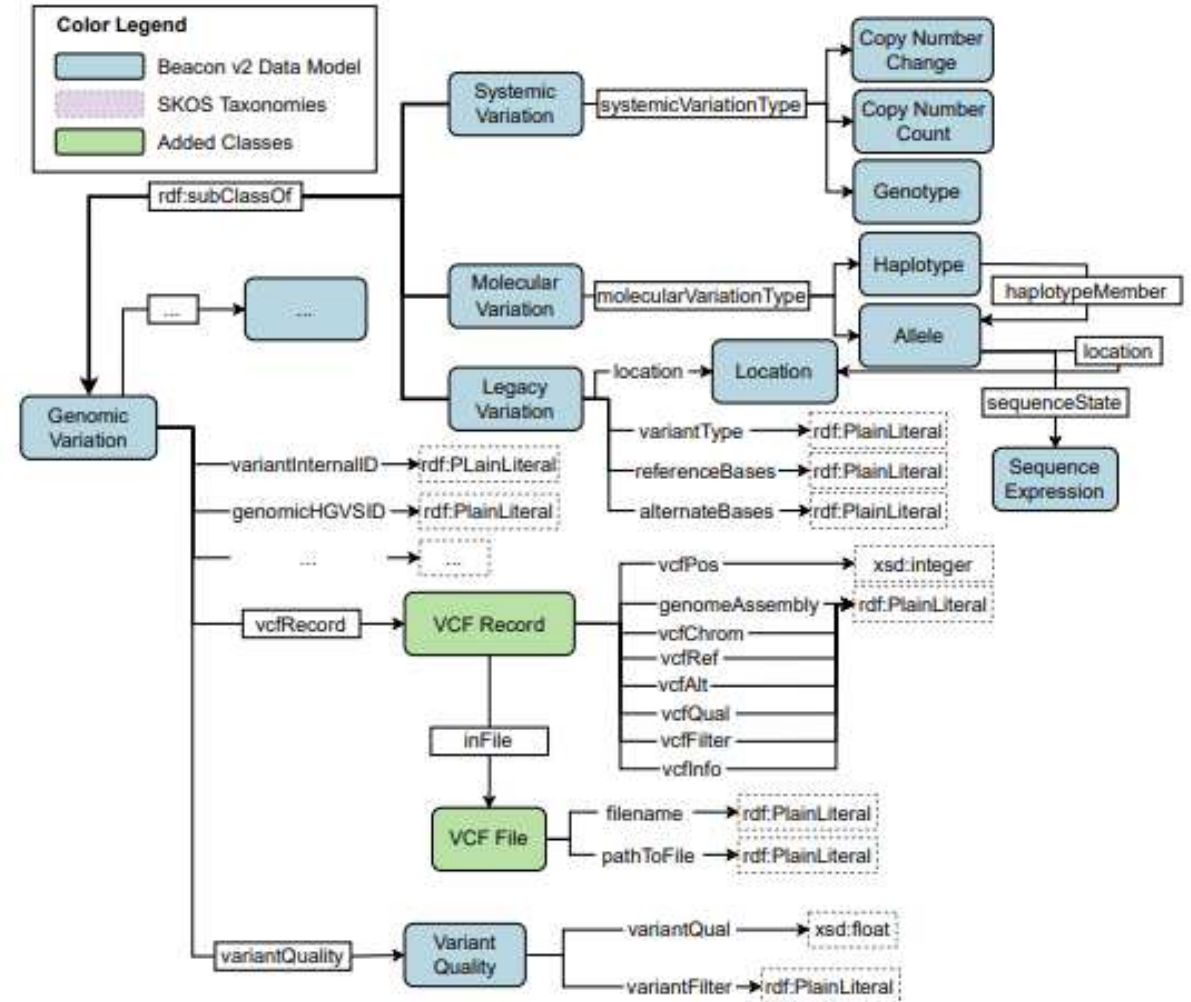
Phased data (G and C above are on the same chromosome)

Representing VCF Semantically?

GA4GH Variant Schema:

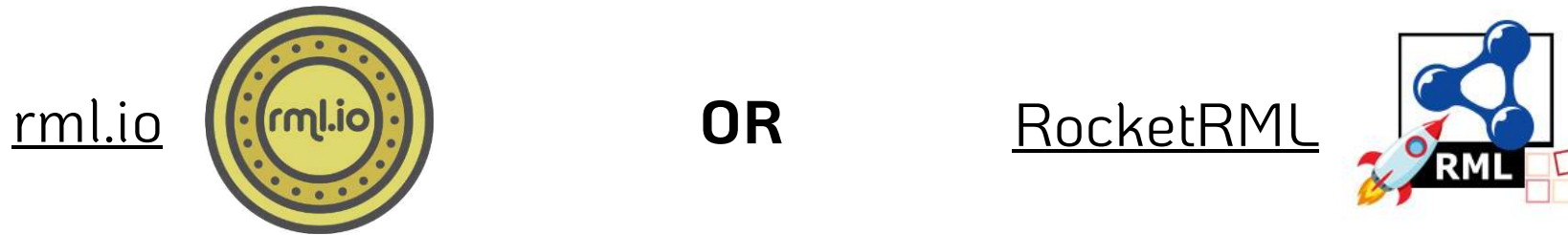


HERO Genomics Ontology:

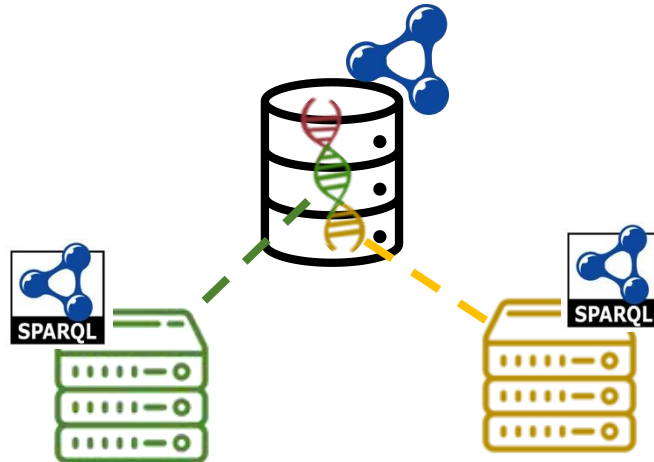


Semantic VCF - Future Aims

Construct VCF Knowledge Graph

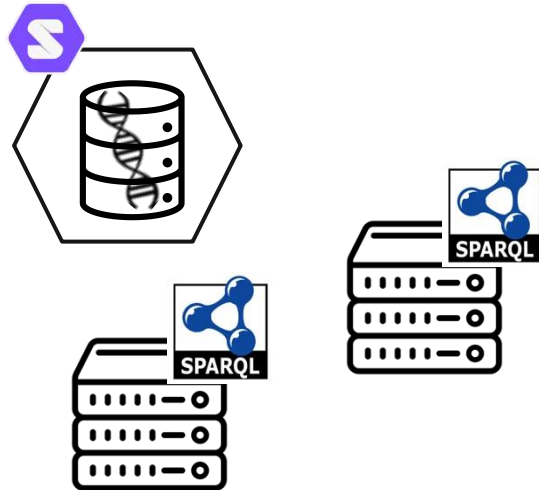


Experiment with SPARQL Querying and Data Linking

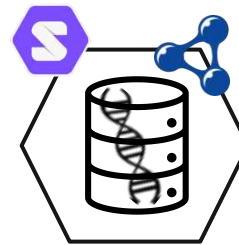


The Third Act – SPARQL Federated Querying

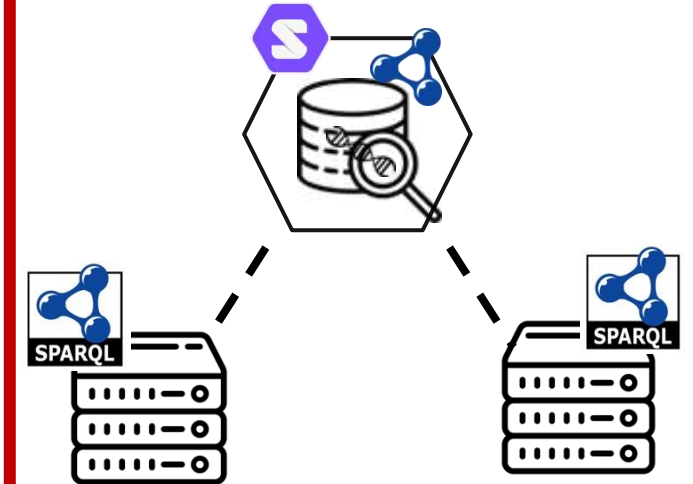
Data Storage



Genomic Data Representation



Question Answering



Motivation:

Federated querying for answering questions over **multiple heterogeneous data sources**

① Performance is currently *~variable~*

↳ We want to detail *WHY* performance varies so much && suggestions for improvement

② Assess federation algorithms on REAL endpoints

↳ **FedX** – uses ASK queries to inform source-selection

↳ **SPLendid** – VOID descriptions to inform source-selection

Real World Federated Query Survey - Design

SPARQL Endpoints:



“Real World” Federated Queries:



<https://sib-swiss.github.io/sparql-examples/>

SPARQL Query Engine:



<https://comunica.dev/>

Example Federated Query

Query: 027-biosodafrontend

Question: (*Bgee* / *UniProt*)

Retrieve the *genes* expressed
in the *human pancreas*

and

those *genes'* *annotations*

```
https://www.bgee.org/sparql/

1 PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
2 PREFIX up: <http://purl.uniprot.org/core/>
3 PREFIX obo: <http://purl.obolibrary.org/obo/>
4 PREFIX genex: <http://purl.org/genex#>
5 PREFIX lscr: <http://purl.org/lscr#>
6 PREFIX orth: <http://purl.org/net/orth#>
7 SELECT DISTINCT ?geneEns ?uniprot ?annotation_text {
8   {
9     SELECT ?geneEns {
10       ?geneB a orth:Gene .
11       ?geneB genex:isExpressedIn ?cond .
12       ?cond genex:hasAnatomicalEntity ?anat .
13       ?geneB lscr:xrefEnsemblGene ?geneEns .
14       ?anat rdfs:label 'pancreas' .
15       ?geneB orth:organism ?o .
16       ?o obo:RO_0002162 ?taxon2 .
17       ?taxon2 up:commonName 'human' .
18     } LIMIT 100
19   }
20 SERVICE <https://sparql.uniprot.org/sparql> {
21   ?uniprot rdfs:seeAlso / up:transcribedFrom ?geneEns .
22   ?uniprot up:annotation ?annotation .
23   ?annotation rdfs:comment ?annotation_text .
24 }
25 }
```

Example obtained from:

https://biosoda.expasy.org/build_biosodafrontend/

visualization from: <https://sib-swiss.github.io/sparql-editor/>

Real World Federated Query Survey – *[Preliminary] Findings 1*

Preliminary Results (endpoints and queries):

Federated querying is not "easy"

↳ Errors related to SPARQL endpoint availability

Invalid SPARQL endpoint response from <https://semantic.eea.europa.eu/sparql> (HTTP status 404)

↳ Errors caused by SPARQL endpoint-related errors

Invalid SPARQL endpoint response from <https://sparql.orthodb.org/sparql> (HTTP status 500)

↳ Errors related to SPARQL endpoint rate-limiting

Invalid SPARQL endpoint response from <https://query.wikidata.org/sparql> (HTTP status 429): Rate limit exceeded

↳ Errors related to SPARQL endpoint time-outs

Unexpected ""!"" at position 0 in state STOP

Real World Federated Query Survey – [Preliminary] Findings 2

Preliminary Results (algorithms):

Federated querying is not "easy"

↳ Errors within Comunica

Actor urn:comunica:default:query-operation/actors#source requires an operation with source annotation.

fetch failed

Could not retrieve <https://idsm.elixir-czech.cz/sparql/endpoint/chebi> (HTTP status 400)

FATAL ERROR: Ineffective mark-compacts near heap limit Allocation failed - JavaScript heap out of memory

↳ Errors related to algorithms

Actor urn:comunica:default:/actors#rdf-join cannot complete join with Asym-hash actor.

Real World Federated Query Survey – *Future Work*

Future Aims:

Execute 1 month worth of queries

↳ 4 experimental runs per week

Vary time + part of week experiment is performed

Publication goals:

- In-Use Paper
- Journal Paper Extension (benchmark creation)

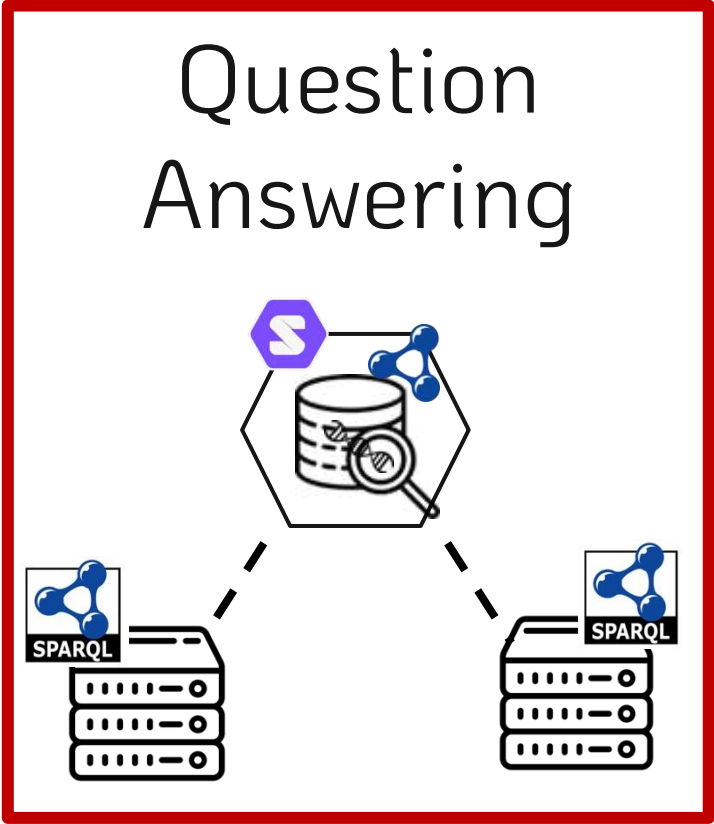
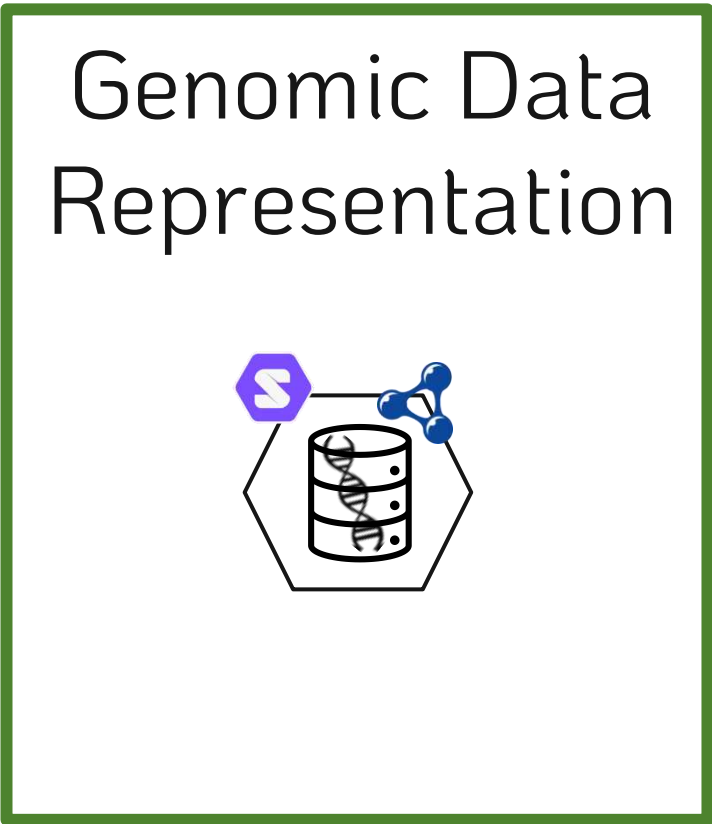
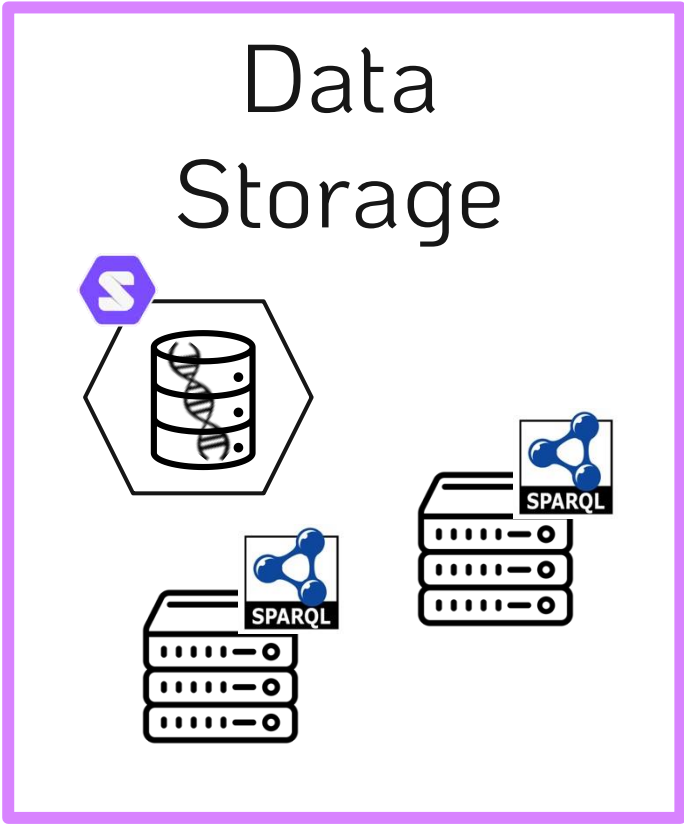
Algorithmic variation

- VOID vs ASK queries
- Various join algorithm combinations
- Rate-limiting
- SERVICE removal

Metrics Assessed

- Total execution time
- Query planning time
- Result set consistency
- # SPARQL requests
- # HTTP requests
- Individual query arrival times

The Finale – Unified System for Storage & Representation & Querying



Framework
& web application



Thank you!



More PhD specifics

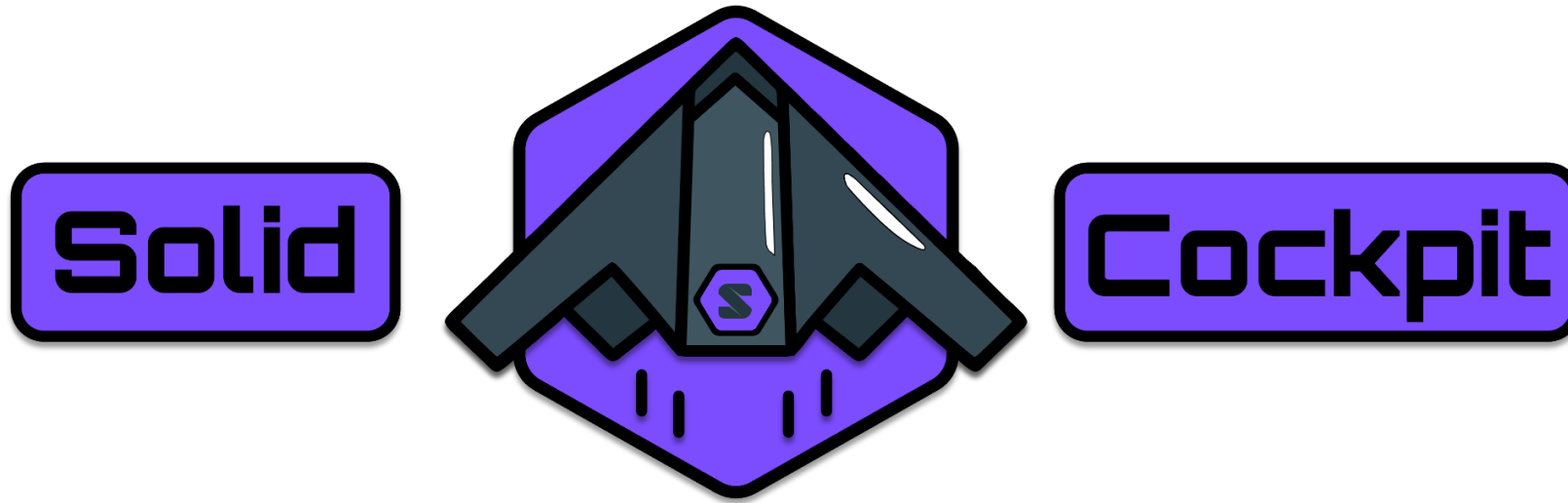


LinkedIn



GitHub

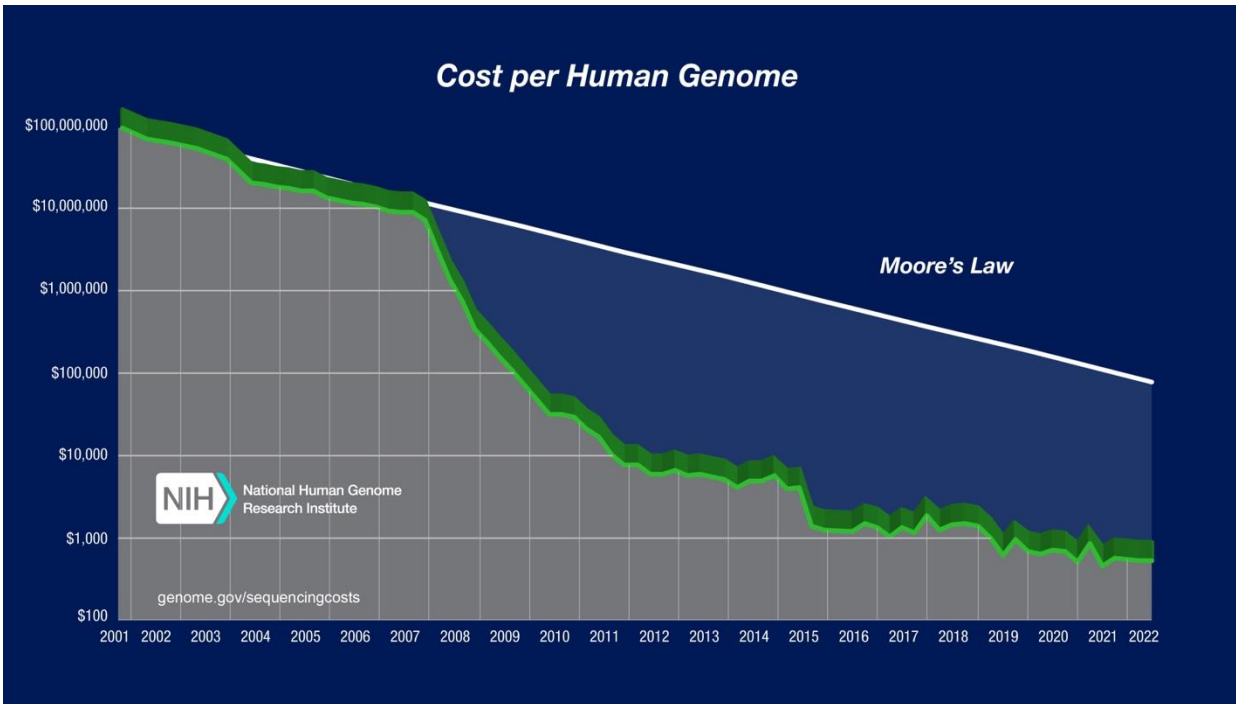
Supplemental Slides



knowledgeonwebscale.github.io/solid-cockpit/

Genome Sequencing in Health Care

Sequence Generation Costs



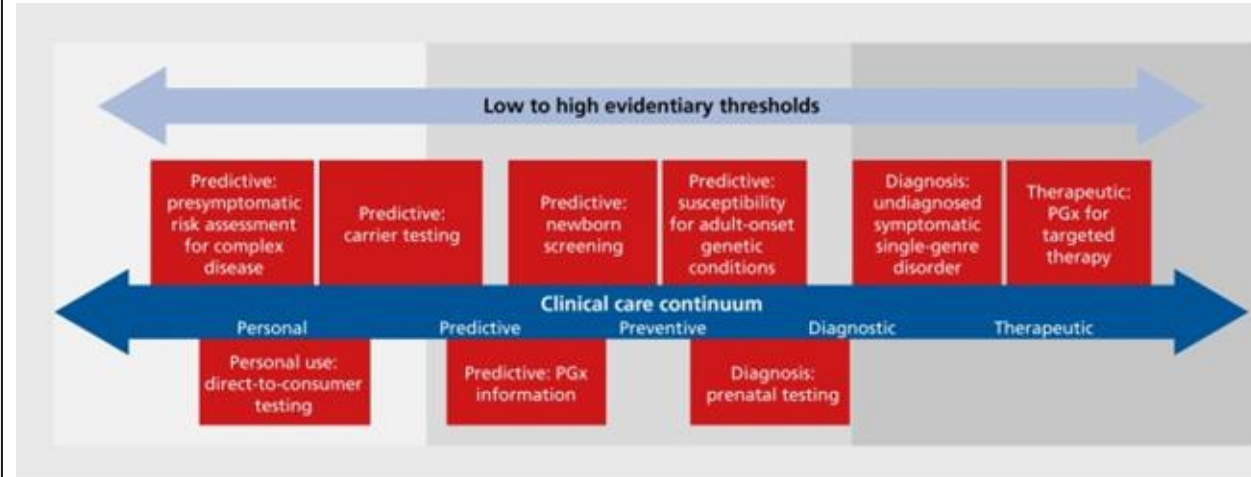
Wetterstrand KA. Accessed 12/04/24.

Currently getting close to \$100 per genome⁴

Per-base cost of sequencing has been dropping by half every ~5 months

Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Variant Call Format (VCF)

- tab delimited .txt file
- Rows represent individual mutations in a sequence (when compared to reference)
- Columns store information about that mutation

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT
--------	-----	----	-----	-----	------	--------	------	--------



```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file:///23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT GEN
chr1 82154 rs4477212 a . . . . GT 0/0
chr1 752566 rs3094315 g A . . . . GT 1/1
chr1 752721 rs3131972 A G . . . . GT 1/1
chr1 798959 rs11240777 g . . . . GT 0/0
chr1 800007 rs6681049 T C . . . . GT 1/1
chr1 838555 rs4970383 c . . . . GT 0/0
chr1 846808 rs4475691 C . . . . GT 0/0
chr1 854250 rs7537756 A . . . . GT 0/0
chr1 861808 rs13302982 A G . . . . GT 1/1
chr1 873558 rs1110052 G T . . . . GT 1/1
chr1 882033 rs2272756 G A . . . . GT 0/1
chr1 888659 rs3748597 T C . . . . GT 1/1
chr1 891945 rs13303106 A G . . . . GT 0/1
```


Personal Genome Sequence Data

Human genome data is big...

~3.2 BILLION base pairs ----> ~1 GB

Sequencing output file ----> ~200 GB

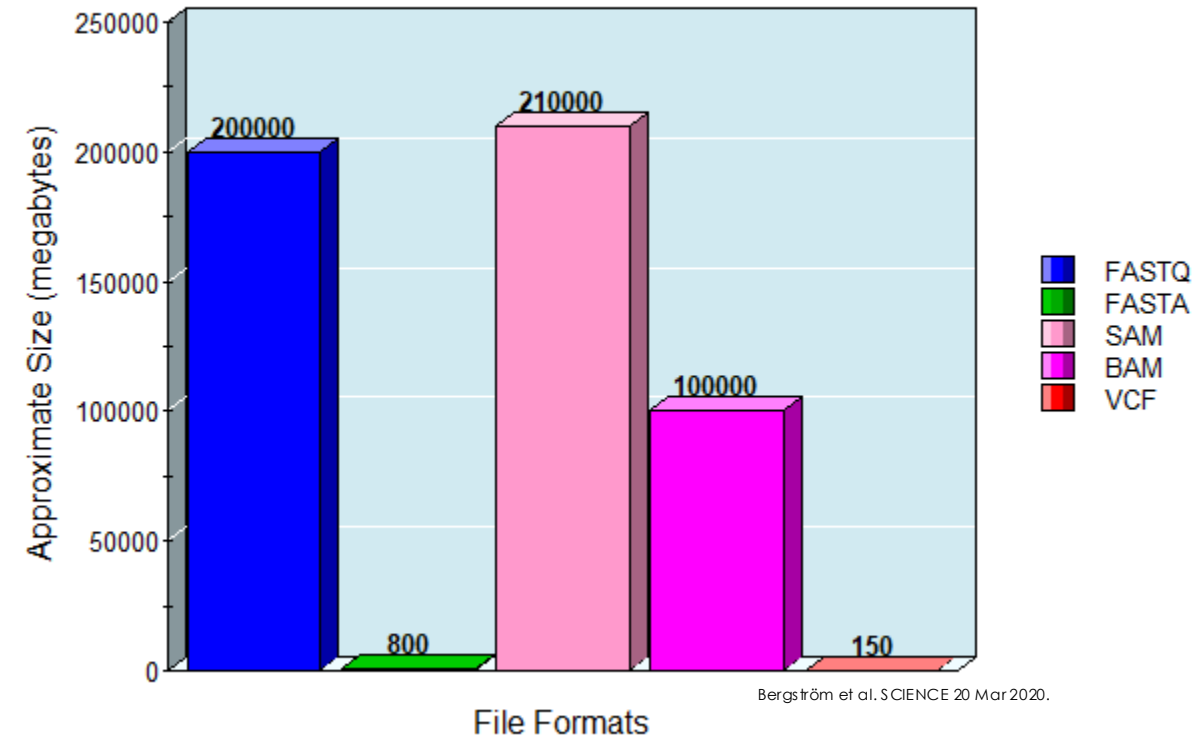
Variant Call Format (VCF) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

VCF file semantic representation

- ~ >**27 MILLION** Triples when represented as RDF
- Ontology for conversion
- Header Dictionary Triples (for compression)

Sequencing File Sizes



Scaling Genome Data Usage

Research genomic data sharing initiatives:



State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces



Not much...

Experimental VCF Data

Primary dataset from publicly available repositories (>100 genomes)

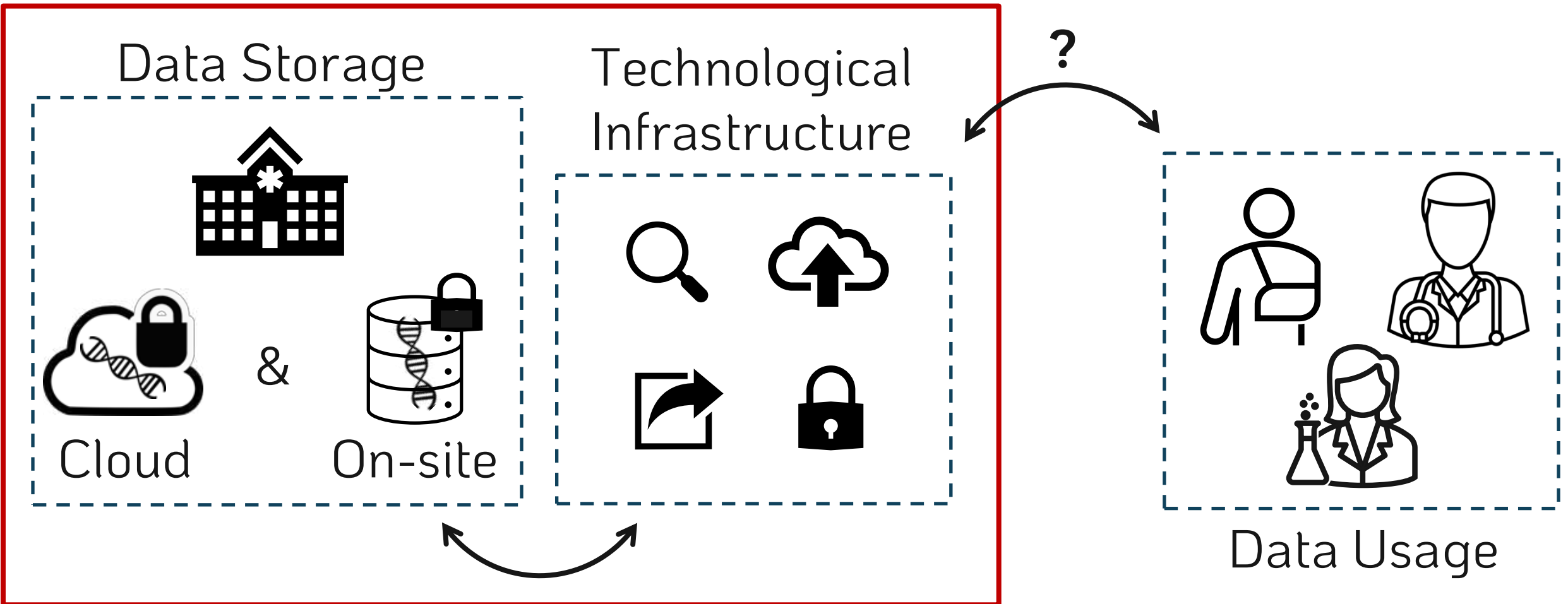
- Personal Genome Project (https://my.pgp-hms.org/public_genetic_data)
- NIST's Genome in a Bottle public dataset (<https://data.nist.gov/od/id/mds2-2336>)
- Illumina Platinum Genomes repository (<https://emea.illumina.com/platinumgenomes.html>)

If more are needed (>1000 [research grade] genomes):

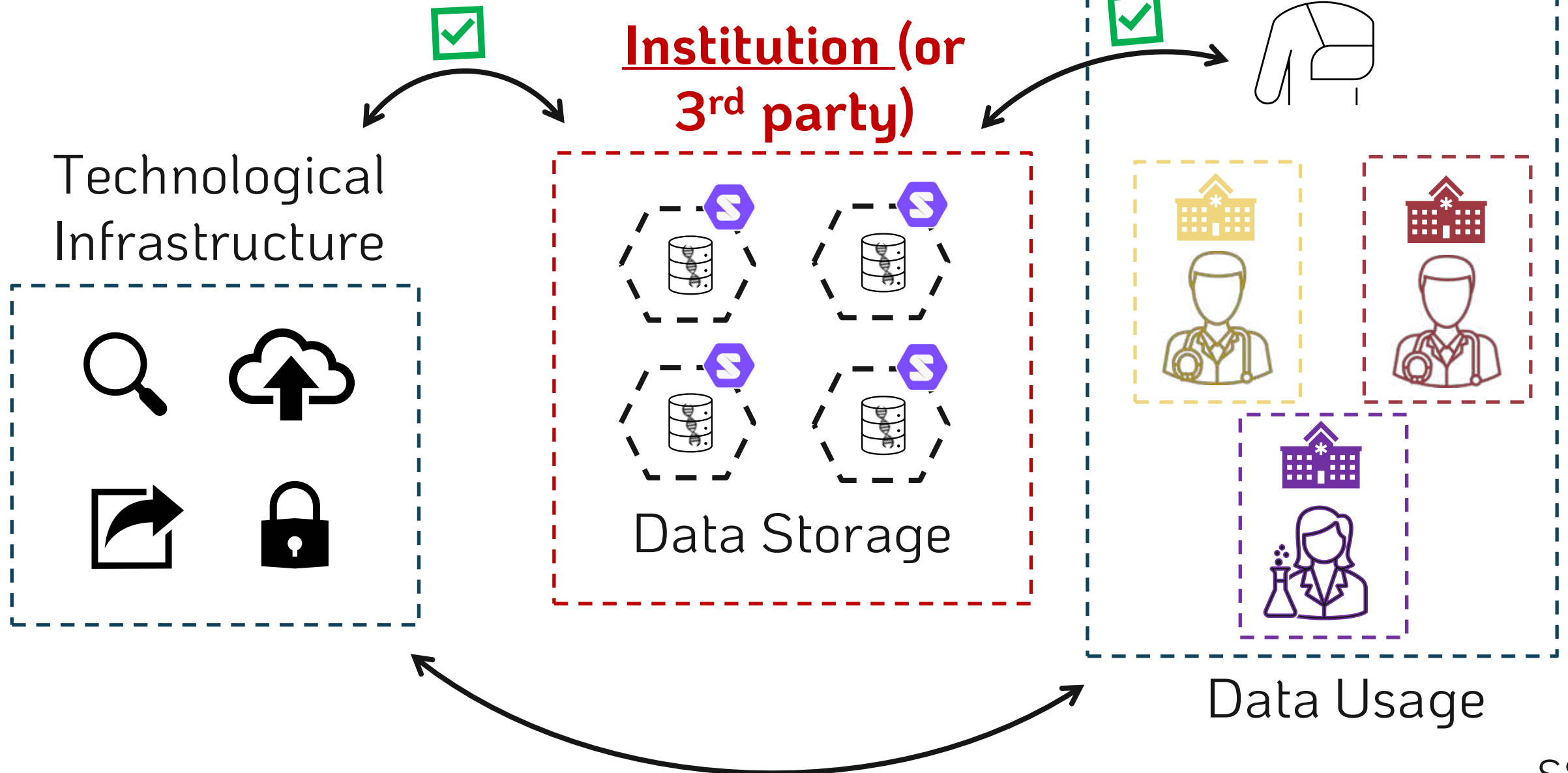
- 1000 genomes project (<https://www.internationalgenome.org/>)
- NCBI's ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>)

Semantic Genome Data?

Institution-centric



A [Hybrid] Solid Solution





- Does not solve ALL problems, but offers an environment where improvements are possible
 - ✓ Easier sharing (web-facilitated)
 - ✓ Granular privacy controls
 - ✓ Direct patient data access
 - ✓ Data representation flexibility (RDF)
 - ✓ Possibility of data-linking and querying

Project Goals



Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

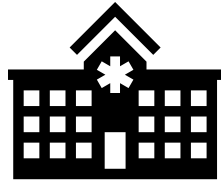
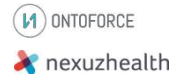
While **exploring representation + querying limits**

Envisioned Product Evolution



PhD End

PhD
Deliverable



Short Term

Start-up / Product
development
+
Hospital implementation



Medical record



Medium Term

Integrate with EHR
+
Increase application
domains

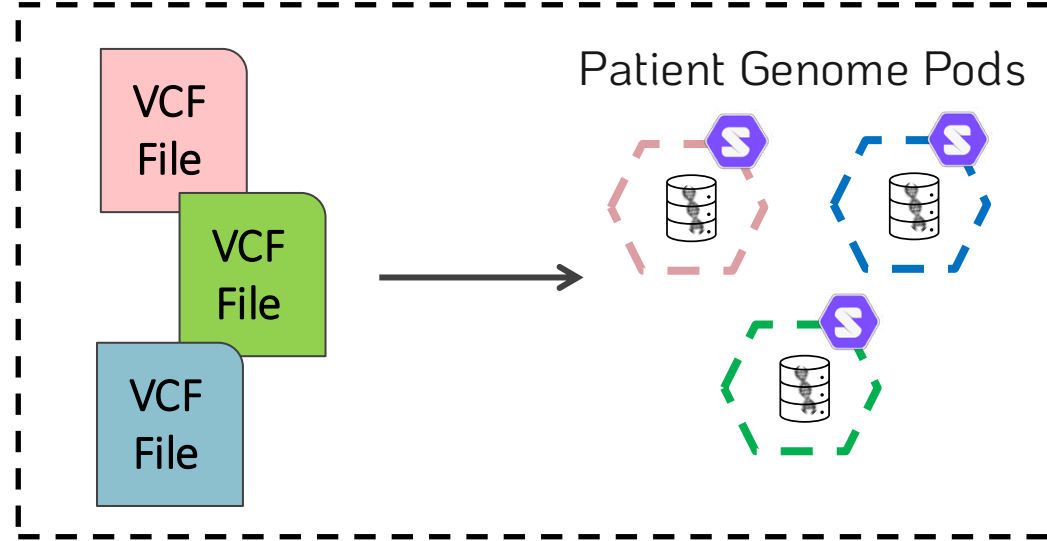


Long Term

Increase 3rd party
application support
+
Scale to Europe and
beyond

Workflow Overview

⚙️ Data Storage



Data Storage



Solid can be implemented for genomic data storage.



Solid pod instances → Community Solid Server¹

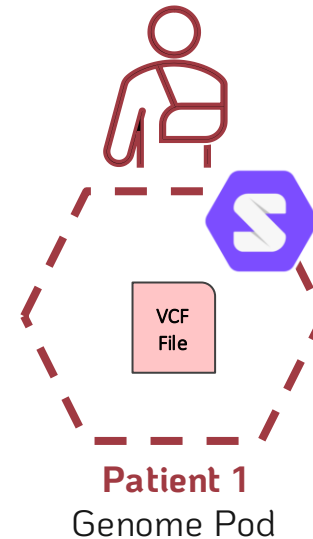


VCF Genomic Data → Illumina Platinum Genomes²

Patient 1
Genome Data

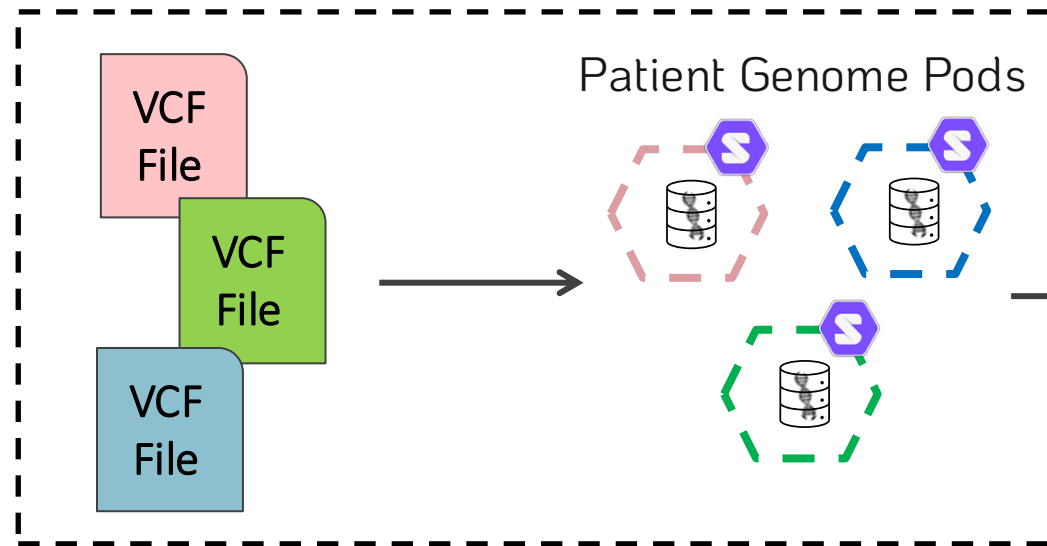


Data Ingestion

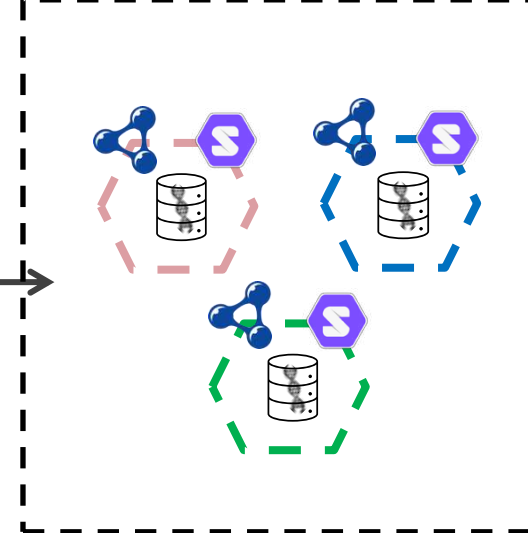


Workflow Overview

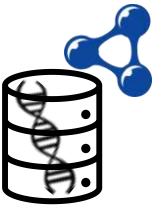
Data Storage



Representation



Genome Data in RDF

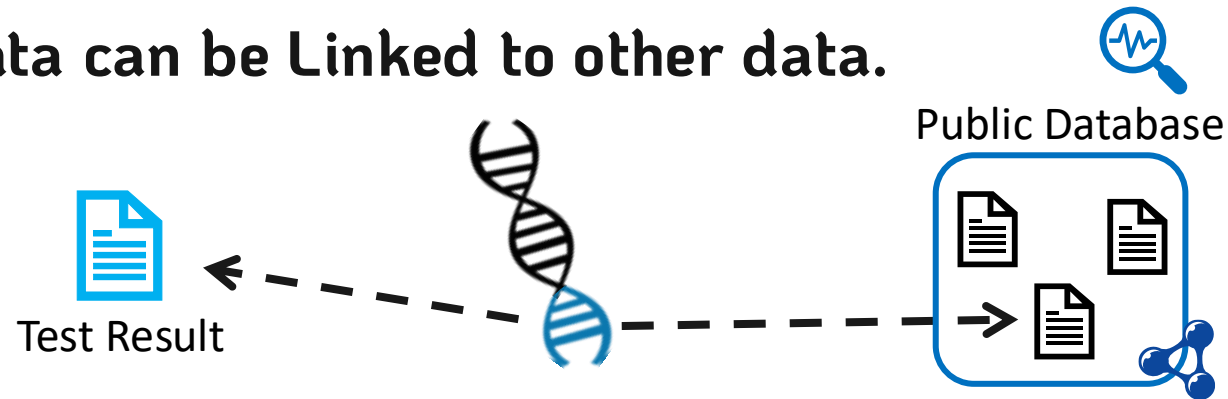


VCF genomic data can be represented as RDF.



 VCF to RDF Vocabularies exist $\longrightarrow$ SPHN Semantic Interoperability Framework⁴
+ FHIR HL7 Format⁵

VCF genomic data can be Linked to other data.



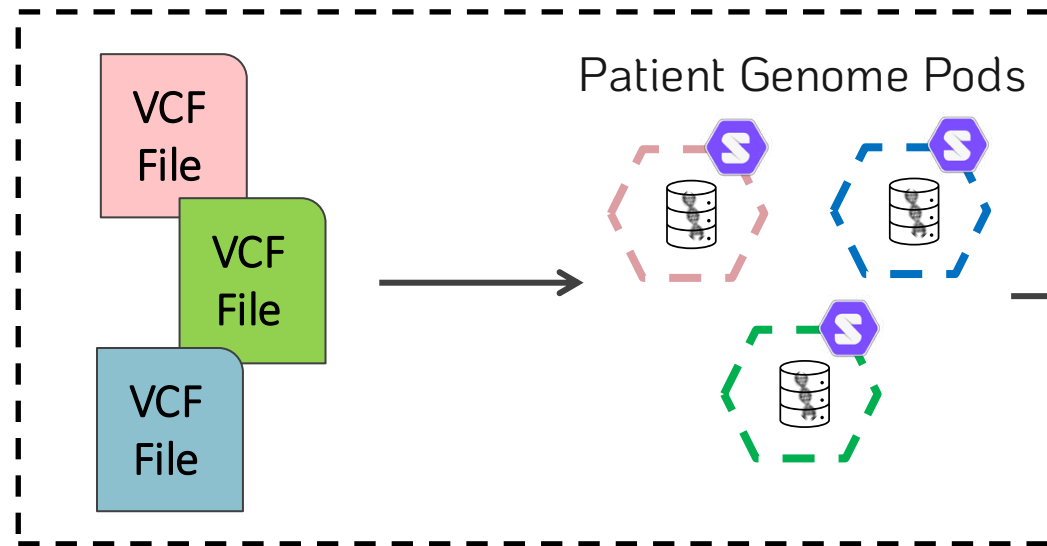
How to automate this?

What vocabularies could be used for the linkages?

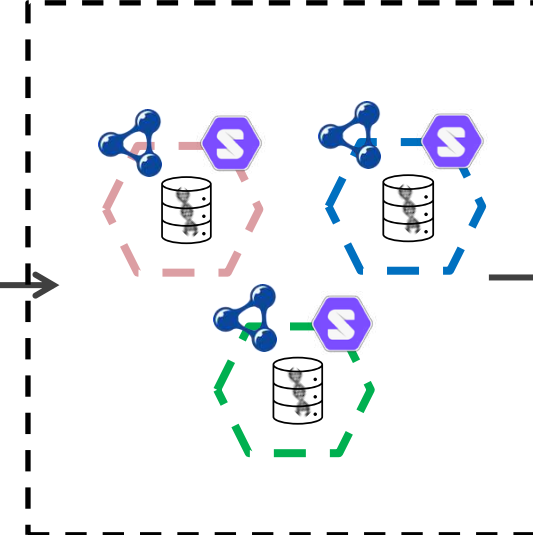


Workflow Overview

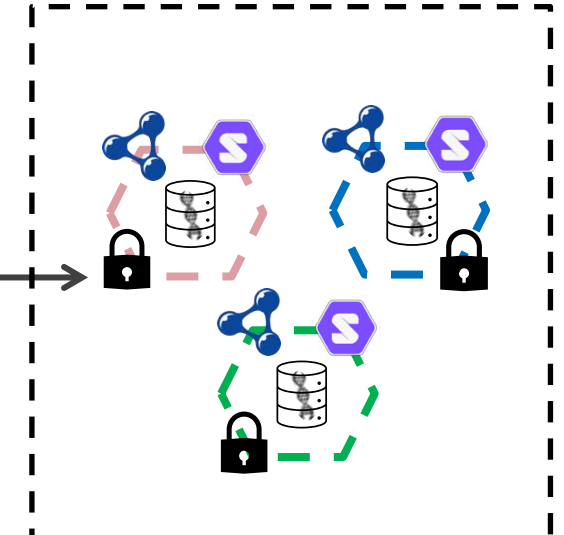
Data Storage



Representation



Privacy



Data Privacy Controls



Granular privacy controls can be implemented in Solid for genomic data.



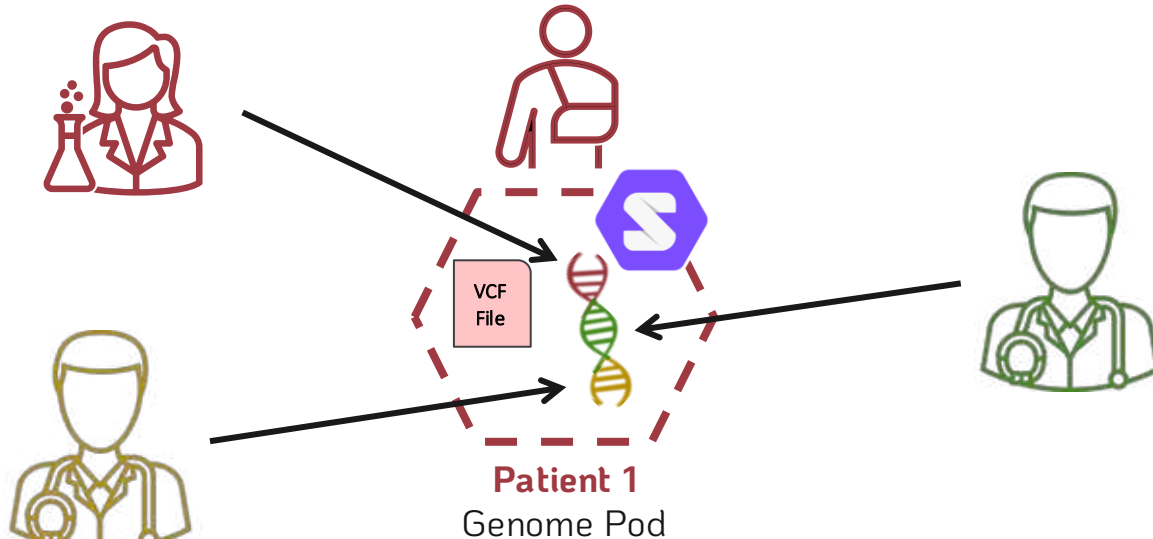
Web Access Control



Solid Protocol ACP³

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

Mutation is also a semantic Class,
above a triple but below a Chromosome

Querying Genome Pods



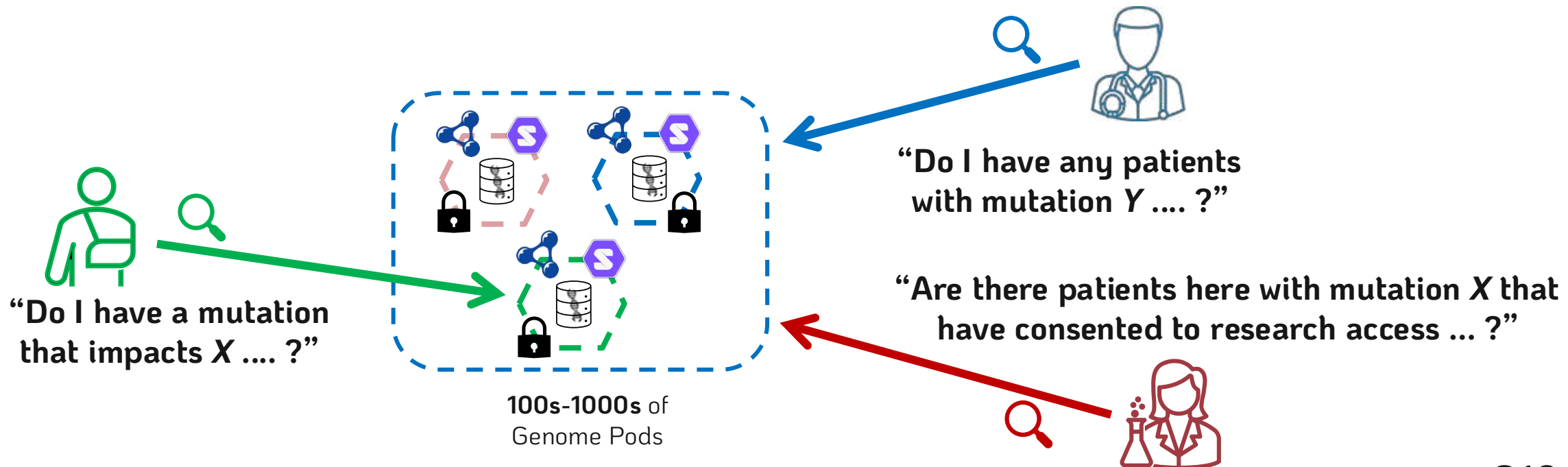
RDF genomic data pods can be queried.



Query Engine



Custom Comunica implementation/engine⁶



⁶ <https://comunica.dev/>

Querying Genome Pods (possibilities)



Single Pod Query

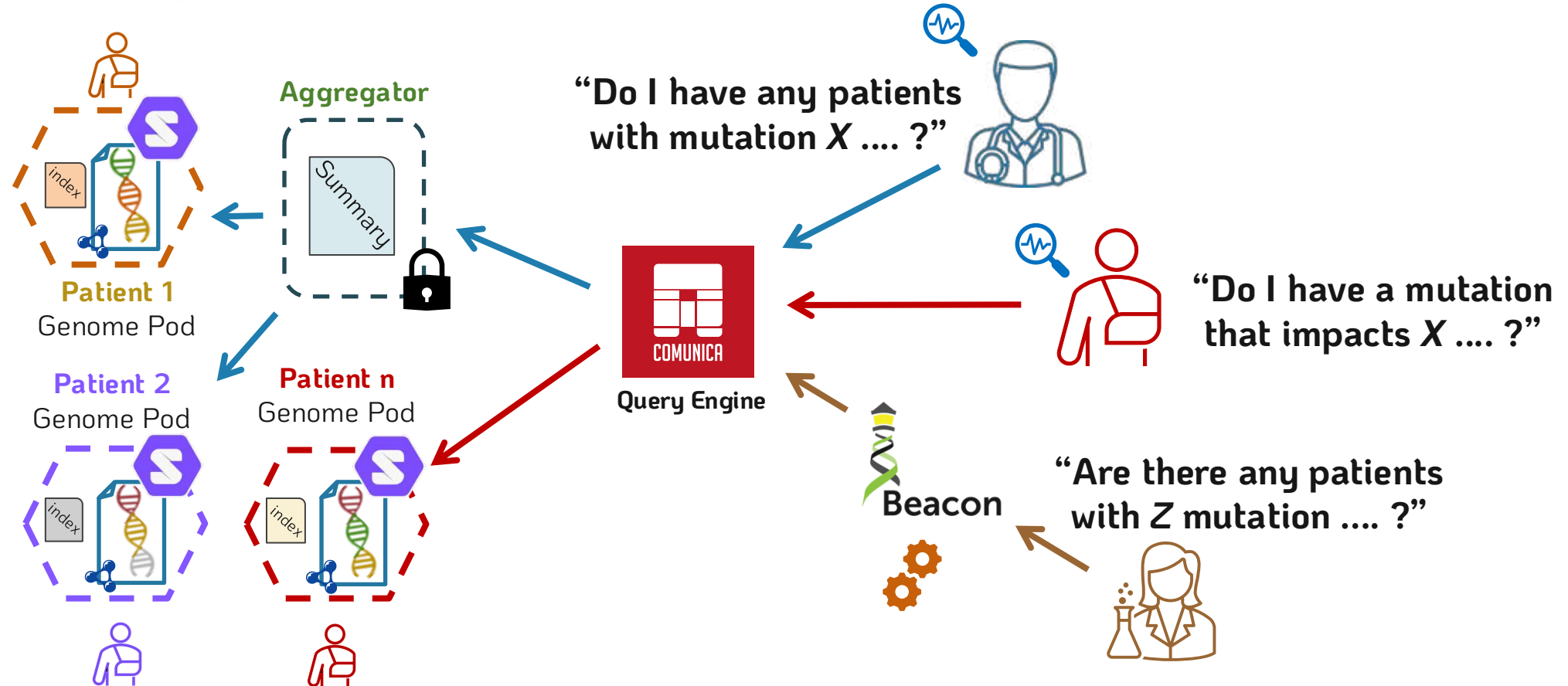
Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed



Implementation Specifics



Solid pod instances → Community Solid Server¹

Genomic Data → Publicly available *VCF* data²

RDF Conversion → HERO Genomics + SPHN ontology⁴

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica⁶